Regularization

High-Dimensional Medical Data

QLS–MiCM Workshop

Why High-Dimensional Data Is Hard

Medical datasets, especially omics, often have:

- **Hundreds–thousands** of predictors
- **Correlated** features
- **Small sample sizes** relative to features

Ordinary linear regression struggles with:

- unstable coefficients
- overfitting
- poor generalization

This motivates regularization.

What Regularization Does

Regularization adds a **penalty** to large coefficients.

This encourages:

- smaller, more stable coefficients
- less sensitivity to noise.
- better generalization on new data

Two main types:

- **Ridge (L2):** shrinks all coefficients
- **Lasso (L1):** sets some coefficients to **zero**

Ridge Regression (L2)

Ridge:

- handles correlated predictors well
- spreads weight across related features
- improves stability without eliminating variables

Key question to explore:

Which features receive the most shrinkage?

Lasso Regression (L1)

Lasso:

- performs **automatic feature selection**.
- drives many coefficients exactly to **zero**.
- useful when features are redundant or weakly informative.

Key question to explore:

Which features does Lasso keep, and why?

Cross-Validation for α (Regularization Strength)

- Splitting your dataset into a training and validation set.
- Cross-validation uses more splits.
- We use cross-validation to estimate regularization strength.